

CTMP — Complete Project Workflow

Document type: Reference workflow diagram pack **Created:** 2026-05-28 (extends `IN_APP_COMPARISON_FLOWCHART_2026-05-27.md`) **Audience:** Anyone onboarding to the project, owner sanity checks, security/compliance reviews

This document covers the **full project workflow end-to-end**, not just the in-app comparison redesign. For the redesign-specific diagrams (PDF viewer flow, Commercial Comparison page layout, etc.), see the 2026-05-27 flowchart — it remains the authoritative source for those slices.

The 10 diagrams below trace a tender from creation through closure, the actors who touch it, the permission gates that guard each surface, and the audit + notification side-effects of every state change.

1. Master swim-lane — the whole tender lifecycle in one picture

```
flowchart TD
    subgraph Owner ["Procurement Admin / Officer"]
        O1[Create tender draft] --> O2[Add criteria + RFQ docs + visibility + invitees]
        O2 --> O3[Submit for approval]
        O3 --> O5[Publish]
        O5 --> O6[Close submissions]
        O6 --> O7[Open technical envelopes]
        O7 --> O9[Finalize technical results]
        O9 --> O10[Schedule committee session]
        O10 --> O11[Open commercial envelopes in session]
        O11 --> O12[Compare + Recommend on Commercial Comparison page]
        O12 --> O13[Confirm award — single click]
        O13 --> O14[Generate Award Minutes PDF on demand]
        O14 --> O15{Need to amend?}
        O15 -- yes --> O16[Amend award — new row supersedes]
        O15 -- no --> O17[Issue formal award → Tender Closed]
    end
    subgraph Approver ["Approver / Finance / Legal"]
        A1[Review in Approvals queue] --> A2{Approve?}
        A2 -- yes --> A3[Approved]
        A2 -- no --> A4[Reject → back to Draft]
    end
    subgraph Vendor ["Vendor"]
        V1[Self-register with CAPTCHA] --> V2[Email verify]
        V2 --> V3[Browse PUBLIC + invited tenders]
        V3 --> V4[Start bid wizard]
        V4 --> V5[Upload technical PDFs]
        V5 --> V6[Upload commercial PDFs]
        V6 --> V7[Submit bid — immutable]
        V7 --> V8[Receive submission receipt + SHA-256]
        V8 --> V13[Receive award outcome — banner / email if opted in]
    end
    subgraph Evaluator ["Technical Evaluator"]
        E1[Open Technical Evaluation page] --> E2[View Full Proposal via PDF viewer]
        E2 --> E3[Score per criterion]
        E3 --> E4[Mark PASS / FAIL]
    end
    subgraph Committee ["Commercial Committee"]
        C1[Attend scheduled session] --> C2[Mark attendance]
        C2 --> C3[Quorum + Chair check]
        C3 --> C4[Witness envelope opening]
    end
    subgraph System ["CTMP System"]
        S1[Audit-log every state change + view + download]
        S2[Hash-chain audit_logs row]
        S3[Send email notifications]
        S4[Generate Award Minutes PDF via puppeteer]
    end
    O3 --> A1
    A3 --> O5
    A4 --> O1
    V1 --> O1
    V4 --> O7
    V7 --> O9
    V8 --> E1
    E4 --> O12
    C1 --> O10
    C4 --> O11
    S1 --> O1
    S1 --> O3
    S1 --> A1
    S1 --> A3
    S1 --> O5
    S1 --> A4
    S1 --> O1
    S1 --> O5
    S1 --> V3
    S1 --> O7
    S1 --> E1
    S1 --> E4
    S1 --> O9
    S1 --> O11
    S1 --> C2
    S1 --> O17
    S1 --> V13
    S1 --> O5
    S1 --> S3
    S1 --> O11
    S1 --> S1
    S1 --> O13
    S1 --> S1
    S1 --> O13
    S1 --> S3
    S1 --> O14
    S1 --> S4
    S1 --> O17
    S1 --> S3
    S1 --> S1
    S1 --> S2
```

2. Tender state machine — all transitions

stateDiagram-v2 [*] --> Draft : tender:create Draft --> InternalReview : submit-for-approval
InternalReview --> Draft : reject (with reason) InternalReview --> Approved : approve Approved
--> Published : publish
(requires procurementType +
estimatedBudget + ≥1 RFQ doc +
≥3 invitees if INVITATION_ONLY) Published --> ClarificationPeriod : clarifications opened
Published --> SubmissionClosed : close-submissions ClarificationPeriod --> SubmissionClosed
: close-submissions SubmissionClosed --> TechnicalOpening : open technical envelopes
TechnicalOpening --> TechnicalEvaluation : evaluators begin scoring TechnicalEvaluation -->
CommercialSealed : finalize technical results
(only PASS vendors proceed) CommercialSealed --> CommitteeCommercialOpening :
committee session opens envelopes CommitteeCommercialOpening -->
CommercialEvaluation : envelopes OPENED CommercialEvaluation -->
AwardRecommendation : recommend (legacy 2-step) CommercialEvaluation --> Awarded :
Confirm (Phase D single-step) AwardRecommendation --> Awarded : approve
AwardRecommendation --> CommercialEvaluation : reject Awarded --> Awarded : Amend
Award
(creates new row, supersedes prior) Awarded --> TenderClosed : Issue formal award Draft -->
Cancelled : cancel (with reason) InternalReview --> Cancelled : cancel Approved --> Cancelled :
cancel Published --> Cancelled : cancel ClarificationPeriod --> Cancelled : cancel TenderClosed
--> Archived : manual archive Cancelled --> Archived : manual archive

3. Vendor onboarding — registration → first bid

flowchart TD V0[Vendor visits /register on vn.hadclinic.com.kw:4201] --> V1[Fill form:
company + email + password + contact] V1 --> V2[Solve hCaptcha challenge — server-side
validated] V2 --> V3{CAPTCHA passes?} V3 -- no --> V0 V3 -- yes --> V4[Vendor row created
status=PENDING
VendorUser created status=ACTIVE] V4 --> V5[Verification email sent — token valid 24h] V5 -->
V6[Vendor clicks link → POST /vendor-auth/verify-email] V6 --> V7[Email marked verified] V7 --
> V8[Procurement Admin reviews in Vendor Management] V8 --> V9{Approve?} V9 -- yes -->
V10[Vendor status = APPROVED] V9 -- no --> V11[Vendor status = REJECTED] V10 -->
V12[Vendor can now log in + browse + bid] V12 --> V13[Browse /tenders — sees PUBLIC
+ INVITATION_ONLY where invited] V11 -.-> V99[End] classDef gate
fill:#fff3e0,stroke:#ff9800,stroke-width:2px class V3,V8,V9 gate

4. Tender creation → publication

flowchart TD
 T0[Admin clicks New Tender] --> T1[Fill basics:
title + dept + submission deadline]
 T1 --> T2[Add optional:
category + procurementType + KWD budget + visibility]
 T2 --> T3[Save as Draft]
 T3 --> T4[Tender detail page]
 T4 --> T5[Upload RFQ documents — PDF/Office, ≤50MB, server SHA-256]
 T5 --> T6{Visibility?}
 T6 -- PUBLIC --> T8[Configure technical criteria
(library or custom, weights = 100%)]
 T6 -- INVITATION_ONLY --> T7[Manage Invited Vendors panel
add ≥3 vendors]
 T7 --> T8
 T8 --> T9[Submit for Approval]
 T9 --> T10[Status: Internal Review]
 T10 --> T11[Approval workflow — Approver acts in Approvals queue]
 T11 --> T12{Approved?}
 T12 -- yes --> T13[Status: Approved]
 T12 -- no --> T14[Status: Draft with rejection reason]
 T13 --> T15[Admin clicks Publish]
 T15 --> T16{Pre-publish gates pass?}
 T16 --> T17[400: missing fields enumerated]
 T17 --> T16
 T16 -- yes --> T18[Status: Published
+ email vendors (deferred — BUG-016)]
 T18 --> T16

- procurementType set
- estimatedBudget set
- ≥1 RFQ document
- INVITATION_ONLY: ≥3 invitees}

T16 -- no --> T17[400: missing fields enumerated]
 T17 --> T16
 T16 -- yes --> T18[Status: Published
+ email vendors (deferred — BUG-016)]
 T18 --> T16

classDef gate fill:#fff3e0,stroke:#ff9800,stroke-width:2px
 class T6,T12,T16 gate

5. Bid submission — vendor side

flowchart TD
 B0[Vendor opens published tender] --> B1{Bidding allowed?}
 B1 -- "tender not Published/Clarification" --> B2[Banner: Tender submission has closed / cancelled / awarded]
 B1 -- "tender is INVITATION_ONLY + not invited" --> B3[Banner: Invitation only]
 B1 -- yes --> B4[Click START BID]
 B4 --> B5[POST /tenders/:id/bids/draft → Bid + 2 envelopes DRAFT]
 B5 --> B6[Bid wizard: technical envelope]
 B6 --> B7[Upload technical PDFs
client: mime+name check
server: mime+magic bytes+SHA-256]
 B7 --> B8[Bid wizard: commercial envelope]
 B8 --> B9[Upload commercial PDFs
same gates]
 B9 --> B10[Review summary]
 B10 --> B11[Click Submit]
 B11 --> B12{Submit gates pass?}
 B12 --> B13[Error: which gate failed]
 B12 -- yes --> B14[Atomic transaction:
• Bid → SUBMITTED
• Both envelopes → SUBMITTED + LOCKED
• Documents lockedAt set
• Receipt with SHA-256 created]
 B14 --> B15[Vendor sees submission receipt
receiptNumber + receiptHash + per-doc SHA-256]
 B15 --> B16[Bids are IMMUTABLE — no revision after submit]

- both envelopes non-empty
- deadline not passed
- vendor APPROVED}

B12 -- no --> B13[Error: which gate failed]
 B12 -- yes --> B14[Atomic transaction:
• Bid → SUBMITTED
• Both envelopes → SUBMITTED + LOCKED
• Documents lockedAt set
• Receipt with SHA-256 created]
 B14 --> B15[Vendor sees submission receipt
receiptNumber + receiptHash + per-doc SHA-256]
 B15 --> B16[Bids are IMMUTABLE — no revision after submit]

classDef gate fill:#fff3e0,stroke:#ff9800,stroke-width:2px
 class B1,B12 gate

6. Technical evaluation

flowchart TD
TE0[Submission deadline passed] --> TE1[Admin: Open Technical Envelopes]
TE1 --> TE2[Tender → Technical Opening
technical envelopes → OPENED]
TE2 --> TE3[Evaluator opens Technical Evaluation page]
TE3 --> TE4[Pick tender + bid]
TE4 --> TE5[Scorecard loads per-tender criteria
via GET /tenders/:id/criteria]
TE5 --> TE6[Click View Full Proposal]
TE6 --> TE7[GET /bids/:id/envelopes/TECHNICAL/documents/:docId/view]
TE7 --> TE8[Audit row written BEFORE stream — no failing-open]
TE8 --> TE9[PDF streams into modal viewer]
TE9 --> TE10[Evaluator scores per criterion
+ toggles gate met / not met]
TE10 --> TE11[Click Save Evaluation]
TE11 --> TE12[TechnicalEvaluation row + per-criterion scores written]
TE12 --> TE13[Repeat for every bid + every evaluator assigned]
TE13 --> TE14[Admin: Finalize Technical Results]
TE14 --> TE15[Server aggregates:
• if ALL mandatory gates passed → PASS
• else → FAIL
(score is for ranking only — master plan §C4)]
TE15 --> TE16[Tender → Commercial Sealed bid.technicalResult locked]
classDef gate fill:#fff3e0,stroke:#ff9800,stroke-width:2px
class TE15 gate
classDef audit fill:#e3f2fd,stroke:#1976d2,stroke-width:2px
class TE8 audit

7. Committee commercial opening + comparison + Confirm

flowchart TD CO0[Procurement Admin schedules committee session] --> CO1[Adds members + marks Chair] CO1 --> CO2[Session date arrives — physical meeting room] CO2 --> CO3[Procurement Admin opens Committee & Commercial page] CO3 --> CO4[Mark attendance per member: Present / Absent] CO4 --> CO5{Quorum + Chair present?

- $\geq$ required_quorum_count members PRESENT

• required_role_code member PRESENT} CO5 -- no --> CO6[QuorumStatus chip shows reason: 'Need 2 more + Chair must be present' Confirm DISABLED downstream] CO5 -- yes --> CO7[Click Open Commercial Envelopes] CO7 --> CO8[Commercial envelopes → OPENED tender → Commercial Evaluation/Comparison] CO8 --> CO9[Navigate to /commercial-comparison] CO9 --> CO10[Matrix renders:

- Lowest-PASS highlighted with Award icon
- FAIL grayed but still expandable

• Summary ↔ Itemized toggle] CO10 --> CO11[Expand vendor card — 5 blocks: 1 line items 2 tech detail 3 commercial docs 4 vendor profile 5 Recommend button] CO11 --> CO12[Click Recommend on a vendor] CO12 --> CO13[AwardConfirmDialog opens] CO13 --> CO14{Lowest-PASS or override?} CO14 -- lowest-PASS --> CO15[Zero-friction Confirm no text no PDF required] CO14 -- override --> CO16[Required: $\geq$ 100 chars text + PDF upload first POST /award/justification-document] CO16 --> CO17[Returns documentId valid 15 min] CO17 --> CO15 CO15 --> CO18[Toggle vendor notification opt-ins (default OFF for both winner + losers)] CO18 --> CO19[Click Confirm] CO19 --> CO20{POST /award/confirm validates:

- quorum still met
- server recomputes lowestPassBidId
- schema CHECK: override → text+PDF

• bid technicalResult = PASS} CO20 -- fail --> CO21[400 with specific reason] CO21 -.→ CO13 CO20 -- pass --> CO22[Atomic transaction:

- Award row created
- tender → Awarded

• winning bid → AWARDED] CO22 --> CO23[Audit log AWARD_CONFIRMED at CRITICAL] CO22 --> CO24[Best-effort notification dispatch if opt-ins TRUE] classDef gate fill:#fff3e0,stroke:#ff9800,stroke-width:2px class CO5,CO14,CO20 gate classDef audit fill:#e3f2fd,stroke:#1976d2,stroke-width:2px class CO23 audit

8. Post-Confirm — notifications + Minutes + amendment

flowchart TD
 PA0[Tender → Awarded] --> PA1{notify_winner flag?}
 PA1 -- TRUE --> PA2[Resolve winning vendor primary contact email
fall back to all active vendor users]
 PA2 --> PA3[Send TENDER_AWARDED_WINNER template
via NotificationsService.sendEmail]
 PA3 --> PA4[NotificationLog row written
status=SENT or FAILED]
 PA4 --> PA5{notify_losers flag?}
 PA5 -- TRUE --> PA6[For each losing bid:
send TENDER_AWARDED_LOSER]
 PA5 -- FALSE --> PA7
 PA6 --> PA7[Audit AWARD_NOTIFICATIONS_DISPATCHED
outcomes list preserved in metadata]
 PA7 --> PA8[Forgot opt-in?]
 PA8 --> PA1
 PA0 --> PB1[Admin clicks Generate Award Minutes]
 PB1 --> PB2[GET /award/minutes.pdf]
 PB2 --> PB3[AwardMinutesService.generate:

- Aggregate tender + award + bids + attendance
- Render HTML template
- puppeteer-core + system chromium → PDF
 - SHA-256 hash + MinIO storage

]
 PB3 --> PB4[Audit AWARD_MINUTES_GENERATED]
 PB4 --> PB5[Stream application/pdf inline
+ X-Award-Minutes-Sha256 header]
 PB5 --> PB6[Admin downloads
award-minutes-.pdf]
 PB6 --> PB7[Re-clicking always generates a NEW row + file
per master plan H2]
 PA0 --> PC1{Need to amend?}
 PC1 -- yes --> PC2[Admin clicks Amend
Award on tender detail]
 PC2 --> PC3[AmendAwardDialog opens]
 PC3 --> PC4[Required: pick
new PASS bid
+ ≥100 char reason
+ PDF upload]
 PC4 --> PC5[POST /award/amend]
 PC5 --> PC6[Atomic transaction:

- new Award row
- prior row superseded_by_award_id set
 - awardedVendorId updated
 - prior winning bid → SUBMITTED

]
 PC6 --> PC7[Audit AWARD_AMENDED at CRITICAL]
 PC7 --> PC8[Both original + amendment remain visible forever
per master plan F7]
 classDef audit fill:#e3f2fd,stroke:#1976d2,stroke-width:2px
 class PA4,PA7,PB4,PC7 audit

9. RBAC visibility — who can do what at each stage

```

flowchart LR
    subgraph Roles
        R1[SYSTEM_ADMIN]
        R2[PROCUREMENT_ADMIN]
        R3[PROCUREMENT_OFFICER]
        R4[APPROVER]
        R5[TECHNICAL_EVALUATOR]
        R6[COMMERCIAL_EVALUATOR]
        R7[COMMERCIAL_COMMITTEE_MEMBER]
        R8[AUDITOR]
        R9[VENDOR_ADMIN / VENDOR_USER]
    end
    subgraph Surfaces
        SUR1[Tender CRUD + Publish + Edit criteria]
        SUR2[Approvals queue]
        SUR3[Technical Evaluation page]
        SUR4[Technical Comparison page]
        SUR5[Committee & Commercial page]
        SUR6[Commercial Comparison page]
        SUR7[Amend Award]
        SUR8[Generate Award Minutes]
        SUR9[Audit Log + Security Alerts]
        SUR10[Evaluation Criteria Library]
        SUR11[System Configuration]
        SUR12[Vendor portal: own bids only]
        SUR13[Commercial bid visibility (commercial:view / download)]
    end
    R1 --> SUR9
    R1 --> SUR10
    R1 --> SUR11
    R2 --> SUR1
    R2 --> SUR2
    R2 --> SUR3
    R2 --> SUR4
    R2 --> SUR5
    R2 --> SUR6
    R2 --> SUR7
    R2 --> SUR8
    R2 --> SUR9
    R2 --> SUR10
    R2 --> SUR11
    R2 --> SUR12
    R2 --> SUR13
    R3 --> SUR1
    R3 --> SUR2
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    R4 --> SUR9
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    R9 --> SUR1
    R9 --> SUR2
    R9 --> SUR3
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    R9 --> SUR5
    R9 --> SUR6
    R9 --> SUR7
    R9 --> SUR8
    R9 --> SUR9
    R9 --> SUR10
    R9 --> SUR11
    R9 --> SUR12
    R9 --> SUR13
    %% Critical exclusion
    R1 -.X. SUR13
    R13 -.X. SUR6
    classDef critical fill:#ffebee,stroke:#c62828,stroke-width:2px
    class SUR13 critical

```

Critical exclusion: SYSTEM_ADMIN does NOT receive commercial:view / commercial:download / comparison:commercial:view by default. This is a deliberate separation-of-duties rule enforced from migration 007 + reinforced by every subsequent migration. Master plan §I reaffirms.

10. Document access + audit gate (read-path security model)

```

flowchart TD
    D0[Caller hits a document endpoint] --> D1{Endpoint type?}
    D1 -- "bid document VIEW" --> D2[GET /bids/:id/envelopes/:type/documents/:docId/view]
    D1 -- "bid document DOWNLOAD" --> D3[GET /bids/:id/documents/:docId]
    D1 -- "tender RFQ document" --> D4[GET /tenders/:id/documents/:docId]
    D1 -- "Award Minutes" --> D5[GET /tenders/:id/award/minutes.pdf]
    D2 --> D6[OptionalVendorOrUserGuard]
    D3 --> D6
    D4 --> D6
    D5 --> D6
    D6 --> D7[JwtAuthGuard + tender:view]
    D6 --> D8[JwtAuthGuard + award:minutes:generate]
    D6 --> D9{Caller type?}
    D9 -- vendor --> D10{Owns this bid?}
    D9 -- internal user --> D11{Envelope type?}
    D10 -- no --> D12[403 Not your bid]
    D10 -- yes --> D13[Allow]
    D11 -- TECHNICAL --> D14{Envelope status = OPENED?}
    D11 -- COMMERCIAL --> D15{Has commercial:view AND envelope OPENED?}
    D14 -- no --> D16[403 Not yet opened]
    D14 -- yes --> D13
    D15 -- no --> D17[403 commercial:view required]
    D15 -- yes --> D13
    D13 --> D7
    D13 --> D8
    D13 --> D18[VIEW path only: write audit row]
    D18 --> D19[Insert document_view_log row]
    D19 --> D20{Audit write succeeded?}
    D20 -- no --> D21[500 — fail closed — NO PDF streamed]
    D20 -- yes --> D22[Stream PDF inline]
    D22 --> D23[DOWNLOAD path: Content-Disposition: attachment]
    classDef gate fill:#fff3e0,stroke:#ff9800,stroke-width:2px
    class D10,D14,D15,D20 gate
    classDef audit fill:#e3f2fd,stroke:#1976d2,stroke-width:2px
    class D18,D19 audit
    classDef deny fill:#ffebee,stroke:#c62828,stroke-width:2px
    class D12,D16,D17,D21 deny

```

Cross-references

- **Locked design rules:** `IN_APP_COMPARISON_MASTER_PLAN_2026-05-27.md`
- **Redesign-specific diagrams:** `IN_APP_COMPARISON_FLOWCHART_2026-05-27.md`
- **Spec definitions:** `implementation-spec.md`
- **Lifecycle states (canonical):** `CLAUDE.md` "Domain Vocabulary"
- **Phase completion:** `IN_APP_COMPARISON_TRACKER_2026-05-27.md`
- **Recent code changes:** `agents/handoffs/HANDOVER.md`

End of project workflow pack.